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Working Group VI - Dwarf Galaxies in Cosmological Sims

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Description

- **Science Question:** Simulate and compare a $10^{10} M_{\text{sun}}$ galactic halo across *all* participating codes
- **Description:** This working group will perform a flagship comparison of cosmological zoom-in galaxies provided by Working Group III to identify and fix problems for each code with the common ICs. The first step of the comparison project in a cosmological context will be simulating a dwarf galaxy halo ($10^{10} M_{\text{sun}}$) in a $L_{\text{box}}=5 \text{ Mpc/h}$. As stated in Working Group

III description, a strong isolation criterion will be imposed to select target halos; that is, $3 R_{\text{vir}}$ radius circle of the halo being simulated must not intersect $3 R_{\text{vir}}$ radius circle of any halo with half or more of its mass at $z=0$. This working group is in charge of selecting two dwarf galaxy halos that fulfill these properties from a 512^3 dark matter only run. One with a quiescent mass accretion history and the other with a significant major merger below $z \sim 2$. It is also in charge of running dark matter only simulations with the different codes to test the Lagrange volumes and try to guarantee that there will be no contamination in the final baryon runs. We will also check if, for these runs, the codes converge on basic characteristics such as the final halo mass and its mass accretion history. Specific science questions involving the running phase and analysis phase still to be addressed. [authored by Jose Onorbe]

A mailing list has been created for all people participating in the working group. Anybody interested in joining should write an email to jonorbeb@uci.edu with the subject "Agora WG-VI mailing list" from the email address from which you want to receive the messages mailing.

– **Status:** Candidates selected. Zoom-in dark-matter only tests using Gadget-2 done for the quiescent halo. MUSIC .conf file to generate the dark matter only ICs and baryon ICs for this halo is already available. Once the tests for the other halo finished, its MUSIC .conf file will also be posted here.

General Information

Cosmology

- LambdaCDM based on WMAP7+BAO+H0 (Komatsu et al. 2011)
- $h=0.702$, $\Omega_{\text{M}}=0.272$, $\Omega_{\text{Lambda}}=0.728$, $\Omega_{\text{b}}=0.0455$
- $n_s=0.961$ and $\sigma_8=0.807$

Each of the participating codes will generate large quantities of unprocessed, intermediate data, in the form of "checkpoints" describing the state of the simulation at a given time. We recommend total 201 timesteps: the first 21 timesteps equally spaced in redshift, $\Delta z = 0.5$, from $z=20$ to $z=10$, and the remaining 180 timesteps equally spaced in $\log(\text{expansion parameter})$, $\Delta \log(a) = |\log(1/11)|/180$, from $z=10$ to $z=0$. Plus redshift snapshots at $z = 6.0, 3.0, 2.0, 1.0, 0.5, 0.2, 0.0$ at the very least.

Mass Resolution in $L_{\text{box}} = 5 \text{ Mpc/h}$

DM run	$m_{\text{dm}} (M_{\odot})$
128 ³	6.41E+06
256 ³	8.01E+05
512 ³	1.00E+05
1024 ³	1.25E+04
2048 ³	1.56E+03

DM+BAR run	$m_{\text{dm}} (M_{\odot})$	$m_{\text{gas}} (M_{\odot})$
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128 ³	5.34E+06	1.07E+06
256 ³	6.67E+05	1.34E+05
512 ³	8.34E+04	1.68E+04
1024 ³	1.04E+04	2.09E+03
2048 ³	1.30E+03	262

Selecting Halo Candidates

First a full-box dark matter only simulation of $L_{\text{box}} = 5 \text{ Mpc}/h$ using 512^3 particles was run down to redshift zero. MUSIC was used to generate the ICs. The run was done using GADGET-2. The halo finding was done using the AMIGA HALO FINDER (AHF).

From the full sample at $z=0$, a dwarf galaxy halo mass bin was selected around $1E10 \text{ Mo}$. The mass limits were 20% of the central mass up and down ($\pm 2E9 \text{ Mo}$). This sample includes 21 halos. After this, the isolation criteria was apply to these halos, that is, $3 R_{\text{vir}}$ radius circle of the halo being simulated must not intersect $3 R_{\text{vir}}$ radius circle of any halo with half or more of its mass. Only the ones that satisfied it were selected for the next step (9 out of 21) and the rest were excluded.

The merger histories of these 9 halos were computed in order to select two candidates: one with a quiescent mass accretion history and the other with a significant major merger below $z\sim 2$. The final decision was made also taking into account the lagrange volume of each halo, which is related with the cpu-cost of the zoom simulation, and trying to avoid very extreme values in their properties: V_{max} , spin, concentration,... (see Figures below).

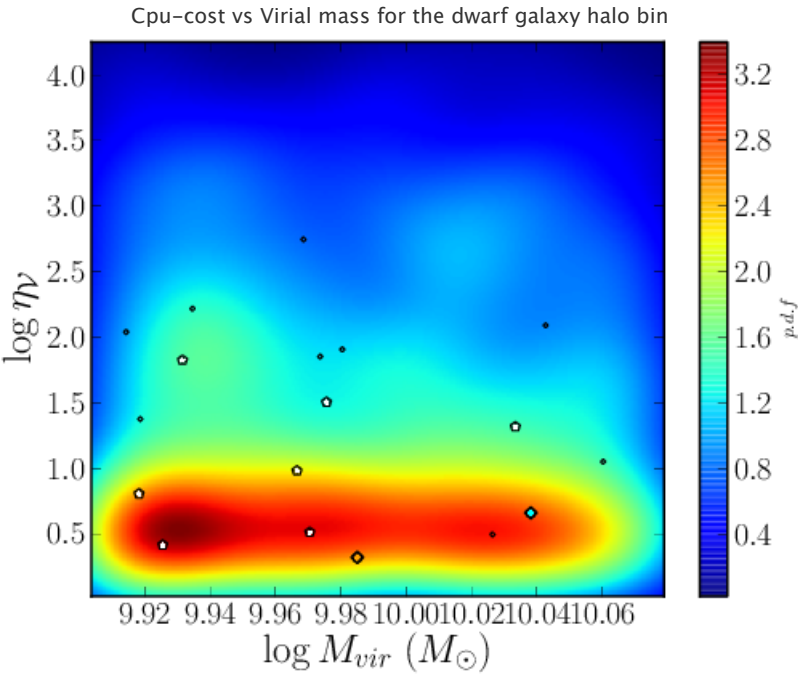


Fig. 1: Efficiency of the Lagrange volume (Number of particles in it compare with the number of particles inside the virial radius) versus virial mass for the dwarf galaxy halo mass bin. Each point is one of the 21 halos in the dwarf galaxy halo mass bin. Black points stand for the halos that did not satisfy the isolation criteria. White points stand for the halos that satisfy the isolation criteria. Cyan and orange points stand for the two halos that satisfy the isolation criteria and were selected as the final candidates (001 and 002 respectively). Color map stands for the probability density function of the same dwarf galaxy mass bin but taken in a box of $25 \text{ Mpc}/h$. The grey band englobes the 25% halos of the sample with the smallest normalized lagrange volume.

Spin vs. Concentration

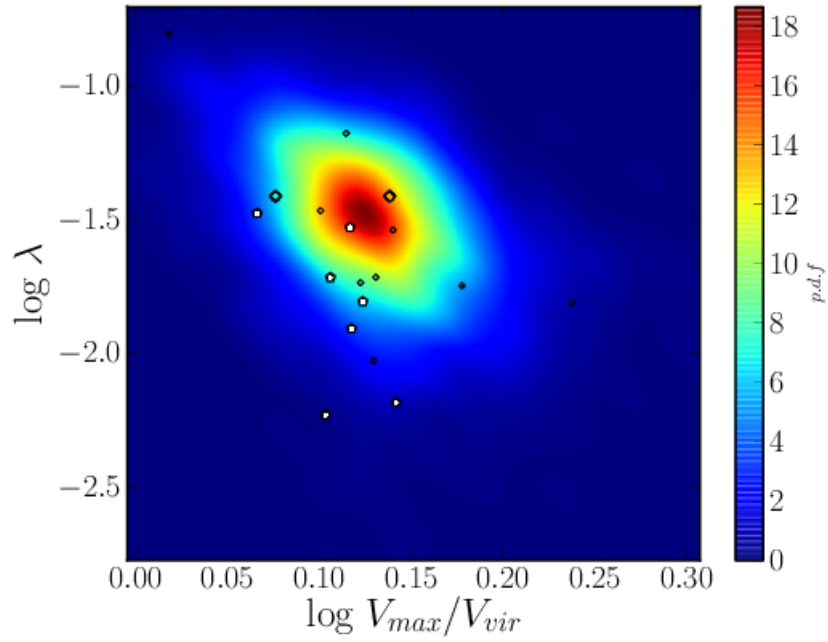


Fig. 2: Spin versus concentration for the dwarf galaxy halo mass bin. Each point is one of the 21 halos in the dwarf galaxy halo mass bin. Black points stand for the halos that did not satisfy the isolation criteria. White points stand for the halos that satisfy the isolation criteria. Cyan and orange points stand for the two halos that satisfy the isolation criteria and were selected as the final candidates (violent and quiescent halos respectively). Color map stands for the probability density function of the same dwarf galaxy mass bin but taken in a box of 25 Mpc/h.

Main Branch Evolution Violent Dwarf Galaxy

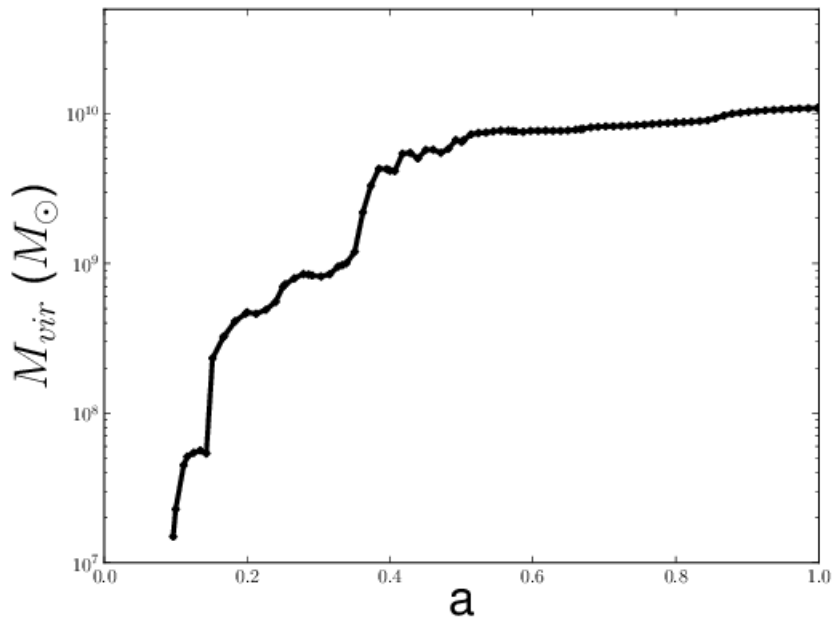


Fig. 3: Main branch evolution for the violent dwarf galaxy (halo 001). This halo has a major merger at $z < 2$.

Main Branch Evolution Quiescent Dwarf Galaxy

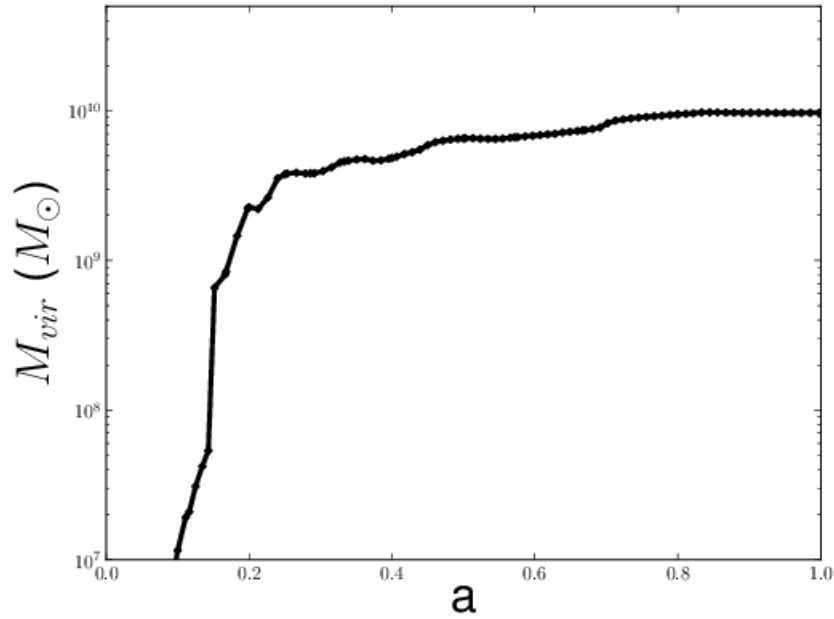


Fig. 4: Main branch evolution for the quiescent dwarf galaxy (halo 002). This halo has a quiescent evolution.

For a more general description on the philosophy proposed in this project regarding the cosmological initial conditions please go to [WG III](#).

Halo Properties and MUSIC .conf Files

Halo	Type	M_{vir} ($M_{\odot}$) ¹	R_{vir} (kpc) ¹	V_{max} (km/s)	zoom 2048 ³ (eff.) MUSIC .conf files ²	M_{200} ($M_{\odot}$) ³	R_{200} (kpc) ³
001	merger	1.09E10	57.8	34.04	file.conf	9.76E9	44.0
002	quiescent	9.63E9	55.5	37.71	file.conf	8.51E9	42.04

¹: To calculate the virial mass and virial radius:

$$\frac{M_{\text{vir}}}{\frac{4\pi}{3}r_{\text{vir}}^3} = \Delta_{\text{vir}}\rho_{\text{(crit/back)}}$$

Δ_{vir} was calculated using spherical top-hat-collapse model. At $z=0$: $\Delta_{\text{vir}} = 360.58$ for $\rho_{\text{background}}$ or $\Delta_{\text{vir}} = 98.08$ for ρ_{crit} .

²: This ascii file is the only input you need to create the ICs of these halos using MUSIC. If you are familiar with MUSIC you should be able to create the ICs for your code from it. However we realize that each code has different particularities and each group has different preferences so please contact with the [Working Group III](#) in case you want more information on the ICs, have any doubts and/or if you want us to create the ICs for you. If you decide to create your own .conf file for your code, please sent a copy to the Working Group VI, so that all the different choices of each group can be tracked (and maybe uploaded here). Finally, the high resolution volume of these specific ICs has been chosen to minimize the cost and guarantee that the halo is not contaminated by low resolution particles. So far, this has been tested by running a dark matter only version of the ICs with GADGET-2.

³: M_{200} and r_{200} were calculated using $\Delta_{\text{vir}} = 200$ and ρ_{crit} .

Participants

Here are the confirmed representatives from participating code groups.

	CODE	VOLUNTEER	1E10q	1E10v
1	ART-Chicago	Sam Leitner		
2	ART-NMSU			
3	ENZO	Christine Simpson		
4	GADGET3(-SPHS)	Miguel Rocha Jose Oñorbe	done (Proj_yt-3.0)	done (Proj_yt-3.0)
5	GADGET3-adaptive force softening			
6	GASOLINE	Sijing Shen		
7	Nyx	Wolfram Shmidt		
8	PKDGRAV2	Mike Kuhlen	done (vis)	
9	RAMSES			

Blog

Working Group VI - Blog

[dwarf002 at z=0 \(pkdgrav2\)](#) Halos identified using Rockstar. Dotted circles are subhalos. Only halos with more than 1000 particles have been plotted.

[edited 11/29/2012: I had an error in my projection routine

Posted Nov 30, 2012, 10:01 AM by Michael Kuhlen

[Untitled Post](#) This is the first blog post for the Working Group VI – Dwarf Galaxies in Cosmological Simulations.

Posted Oct 4, 2012, 12:43 AM by Ji-hoon Kim

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